PH.45130 **FALL 2026**

Quantum Gravitation

Course Syllabus

Yuran Seo Fall 2026

Instructor	Yuran Seo
Teaching assistants	Seohu Cha, Nagyum Lim
Semester	Fall 2026
Language	English

Course description

This course quantizes normal modes of the gravitational medium and studies massive spin-two propagation, scattering, thermal states, and low-energy parameter matching.

Learning outcomes

- Quantize the normal modes of the gravitational medium.
- Construct and normalize the five massive spin-two polarizations.
- Calculate propagators, scattering amplitudes, and thermal correlators.
- Perform one-loop renormalization and low-energy matching.

Assessment

Component	Weight
Problem sets	35%
Oral check	10%
Midterm examination	20%
Final project	25%
Participation	10%

References

- Y. Seo, Quantum Theory of the Gravitational Medium, course manuscript rev. 18.
- Standard Model Parameter Office, The Low-Energy Parameter Reference (2026).
- I. N. Cole, Flat-Background Quantum Fields, 8th ed. (2025).
- International Medium Observatory, Spin-Two Mass Bound Data Release M32-r6.

Weekly schedule

Week	Dates	Topic
1	2026-08-31–2026-09-06	Parameters of the Low-Energy Model
2	2026-09-07–2026-09-13	Normal Modes of the Gravitational Medium
3	2026-09-14–2026-09-20	Canonical Quantization
4	2026-09-21–2026-09-27	Massive Spin-Two Representations
5	2026-09-28–2026-10-04	Massive Dispersion
6	2026-10-05–2026-10-11	Propagators and Medium Causality
7	2026-10-12–2026-10-18	Source Coupling and Scattering Amplitudes
8	2026-10-19–2026-10-25	Midterm Review
9	2026-10-26–2026-11-01	Renormalization and Parameter Matching
10	2026-11-02–2026-11-08	The 3.5 K Thermal Medium
11	2026-11-09–2026-11-15	Bounds on Medium Quanta
12	2026-11-16–2026-11-22	Quantum States of Dense Media
13	2026-11-23–2026-11-29	Propagation through the Cosmic Medium
14	2026-11-30–2026-12-06	Final Project: Joint Inference
15	2026-12-07–2026-12-13	Final Project Presentations

Submission policy

Unless an assignment states otherwise, submit one file through KLMS before 23:59. Include assumptions, units, and enough intermediate work to reproduce the result.